Capital and Labor in Heterogeneous-Agent Economies

Pedro Silos

Temple University

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What We Have Done

- Built a theory of *employment*: where jobs come from, why unemployment exists, how flows move over the cycle, but with risk-neutral workers who never save.
- Learned about *earnings risk*: what the individual income process looks like. It's exogenous, taken as given, faced by households who so far could only save.
- From last semester you are familiar with the Aiyagari machinery: households self-insuring against that risk with a buffer stock, prices clearing in general equilibrium.
- Missing everywhere so far: households who choose **how much to work**, in economies where they also hold **wealth**. That interaction is this topic.

The Questions of This Topic

Four questions organize everything:

1. How does **wealth shape labor supply**—hours and participation—when markets are incomplete?
2. Is there such a thing as *the* **aggregate labor supply elasticity**, and why do micro and macro estimates disagree by an order of magnitude?
3. How does **capital interact with labor on the technology side**—which workers does capital complement, and what does that do to the wage structure?
4. What changes when the labor market that generates the risk meets the households that insure against it?

Five Frameworks

- **The workhorse**: Aiyagari with an hours choice— the mechanics of wealth-dependent labor supply.
- **Pijoan-Mas (2006)**: quantifies the intensive margin of labor as an *insurance technology*.
- **Hansen–Rogerson → time averaging**: the extensive margin, and two ways to smooth an indivisible choice.
- **Chang–Kim (2007)**: aggregation of the extensive margin in stochastic GE—quantities of labor.
- **KORV (2000)**: capital–skill complementarity— *prices* of labor.
- **Krusell–Mukoyama–Şahin (2010)**: frictional labor markets inside the savings economy.

Environment

- A continuum of households; idiosyncratic productivity e follows the a persistent process (Markov chain $\pi(e' \mid e)$); no insurance markets; one riskless asset with limit $a' \geq -\underline{a}$.
- Preferences, our default parameterization:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\mu}}{1-\mu} - B \frac{h_t^{1+1/\nu}}{1+1/\nu} \right].$$

- Firms: $Y = K^\theta L^{1-\theta}$, depreciation δ; L is *effective* labor.
- The household chooses h_t, so earnings $we_t h_t$ are partly a choice.

The Household Problem

Recursively:

$$V(a, e) = \max_{c,\, h,\, a'} \frac{c^{1-\mu}}{1-\mu} - B\,\frac{h^{1+1/\nu}}{1+1/\nu} + \beta \sum_{e'} \pi(e' \mid e)\, V(a', e')$$

subject to

$$c + a' = (1 + r)\, a + w\, e\, h, \qquad a' \geq -\underline{a}, \qquad c, h \geq 0.$$

- Two choices interact: the saving margin and the new **hours margin**.

Optimality Conditions

Multipliers: λ_t on the budget, $\phi_t \geq 0$ on $a_{t+1} \geq -\underline{a}$. First-order conditions:

$$\begin{aligned} c_t: &\quad c_t^{-\mu} = \lambda_t \\ h_t: &\quad B\, h_t^{1/\nu} = \lambda_t\, w\, e_t \\ a_{t+1}: &\quad \lambda_t = \beta(1+r)\,\mathbb{E}_t \lambda_{t+1} + \phi_t \end{aligned}$$

Combining the first and third gives the familiar Euler inequality

$$c_t^{-\mu} \;\geq\; \beta(1+r)\,\mathbb{E}_t\, c_{t+1}^{-\mu}, \quad \text{with equality when } a_{t+1} > -\underline{a}.$$

The Intratemporal Margin

Combining the first two conditions:

$$\underbrace{B\, h_t^{1/\nu}}_{\text{marginal pain of an hour}} = \underbrace{w\, e_t\, c_t^{-\mu}}_{\text{wage}\times\text{marginal utility of consumption}} .$$

- This one equation is the topic's engine. The right-hand side contains c_t, and c_t depends on *wealth*: rich households have low marginal utility, so an hour of work buys them little that they value.
- Hence **hours are decreasing in wealth**, holding productivity fixed (from the FOC).
- Every framework in this part of the course is a different answer to "what does this margin do in equilibrium?"

Why ν Is the Frisch Elasticity

Solve the hours condition: $\ln h = \nu(\ln \lambda + \ln w + \ln e - \ln B)$. Then

$$\left.\frac{\partial \ln h}{\partial \ln w}\right|_{\lambda \text{ fixed}} = \nu.$$

- The **Frisch** elasticity answers: how do hours respond to a wage change *holding the marginal utility of wealth fixed*—i.e., to a temporary, anticipated, or compensated wage movement. The way we parameterize utility the answer is literally the parameter ν.
- It is the natural elasticity for business-cycle questions, where wage movements are transitory and λ barely moves.

Frisch vs. Marshallian: the Units Problem

- The **Marshallian** (uncompensated) response lets λ adjust: a *permanent* wage increase raises lifetime resources, lowers λ, and pulls hours back down—the wealth effect. Marshallian $<$ Frisch, possibly negative.
- Micro studies overwhelmingly exploit *persistent* wage variation (tax reforms, life-cycle profiles): they estimate Marshallian/Hicksian objects, and find them small.
- Macro models need the *Frisch* for fluctuations.
- So part of the famous micro–macro elasticity fight is a **units mismatch** (that is, comparing different theoretical objects). The rest of the debate is real.

Hours and Wealth: the Mechanism

From the intratemporal condition at our default calibration ($\mu = 1$, $\nu = 0.5$, $B = 12$, $w = 1$, $e = 1$, consuming annuity income plus earnings):

assets a	**0**	**2**	**5**	**10**	**20**
hours h	0.44	0.19	0.13	0.09	0.06

- A household near the constraint works two to seven times the hours of a wealthy one, *at the same wage and productivity*.
- Reading: the wealth-poor have a cheap substitute for the buffer stock they lack—**work more**. Labor supply is a second self-insurance margin, strongest exactly where the asset margin is weakest.

A Knife-Edge Worth Knowing

Let $x \equiv (1+r)a - a'$ (net non-labor resources). With log utility, the intratemporal condition plus the budget give

$$\operatorname{sign}\left(\frac{\partial h}{\partial e}\right) = \operatorname{sign}(x).$$

- $x > 0$: substitution dominates—hours *rise* with productivity.
- $x = 0$ (hand to mouth in labor income): substitution and wealth effects cancel **exactly**—balanced-growth preferences at the individual level.
- $x < 0$ (saving heavily out of earnings): hours *fall* with productivity.

Equilibrium Definition

A stationary recursive competitive equilibrium: V, policies (g_a, g_c, g_h), prices (r, w), distribution λ^* s.t.

1. Given (r, w), V solves the Bellman equation and the policies attain it.
2. $r = \theta(K/L)^{\theta-1} - \delta, \quad w = (1-\theta)(K/L)^{\theta}$.
3. λ^* is the fixed point of the law of motion induced by g_a and the chain for e.
4. Markets clear: $K = \int a \, d\lambda^*$ and $L = \int e \, g_h(a, e) \, d\lambda^*$.

Computation

- Outer loop: bisect on r (which pins w through the firm's conditions).
- Inner problem: at each state and candidate a', the intratemporal condition plus the budget deliver (c, h) as the solution of one nonlinear equation in h.
- Then: stationary distribution, aggregation, update r.
- Note that aggregate *effective* labor, $\int e\, g_h \, d\lambda^*$, not raw hours—the classic bug.
- Also, the ergodic asset set is wider than with fixed hours (work-more is a substitute for save-more, but so is the reverse), so make sure grid is large enough.

Taking Stock

- Hours respond to wealth through marginal utility; labor supply is self-insurance; the aggregate response of hours to *anything* therefore depends on the wealth distribution.
- We don't know yet **how much** any of this matters. Is the insurance value of the hours margin quantitatively important? Does it change aggregates? Welfare?
- Those are calibration questions, and answering them carefully is **Pijoan-Mas (2006)**.

Pijoan-Mas (2006)

"Precautionary Savings or Working Longer Hours?" (*RED*).

- Households facing uninsurable risk have *two* self-insurance margins: the buffer stock, and hours. The Aiyagari tradition quantified the first with the second shut down.
- Pijoan-Mas asks: how much insurance does the labor margin provide, who uses it, and what does it do to aggregates and welfare?
- Why it matters: if hours are insurance, then (i) observed hours dispersion partly *is* insurance, not preferences; (ii) aggregate K and H respond to market incompleteness; (iii) the welfare cost of missing markets is mismeasured when hours are held fixed.

The Model

- The model is exactly Part 1's economy—nonseparable consumption–leisure preferences (so wealth and substitution effects are both live), persistent AR(1) productivity, borrowing limit, general equilibrium through K and effective L.
- Calibration: wage process from PSID-style panel estimates ; preferences to match average hours and the capital–output ratio; validated against the joint distribution of wealth, wages, and hours.

The Identifying Design: a 2×2

The experiments that isolate what the labor margin contributes:

	hours flexible	**hours fixed**
incomplete markets	benchmark	shut the margin
complete markets	first-best comparison	RA benchmark

- Compare across the top row: what flexibility does *given* incomplete markets.
- Compare down the first column: what incompleteness does *given* flexibility—including how much of it the labor margin absorbs.

Finding 1: Insurance in the Cross Section

- Conditional on wages, **hours decline with wealth**: the wealth-poor work more—the Part 1 mechanism, alive in the calibrated equilibrium and consistent with the data he matches.
- Consequence for measurement: part of observed hours dispersion across households reflects *positions in the wealth distribution*, not preference heterogeneity.
- Any micro estimation that reads hours differences as tastes—or any calibration that matches hours dispersion with preference shocks—is partly fitting insurance behavior.

Finding 2: Precautionary Labor Supply

- Relative to the complete-markets benchmark, households both **over-save and over-work**: market incompleteness raises aggregate capital *and* aggregate hours.
- The precautionary motive, familiar from 8002 on the asset margin, operates on the labor margin too—and part of it migrates from saving to working, since the two are substitute buffers.
- So "precautionary savings *or* working longer hours" resolves as: both, jointly determined, with the mix varying across the wealth distribution (the poor lean on hours, the comfortable on assets).

Finding 3: Welfare

- Hours flexibility **substitutes for the missing insurance markets**: the welfare cost of market incompleteness is substantially smaller when households can use the labor margin than when hours are fixed.
- Interpretation to retain: evaluating market incompleteness, or any policy that interacts with it, in a fixed-hours model *overstates* the costs borne by households who could have self-insured with work—and *understates* the burden on those who cannot (the sick, the constrained-by-jobs; a door KMŞ walks through).

Lessons from Pijoan-Mas

- The intratemporal margin is a quantitatively serious insurance technology, not a garnish on the Aiyagari model.
- Calibration discipline: preference parameters set in a fixed-hours world are wrong in a flexible-hours world; the insurance role of hours contaminates "labor supply elasticity" estimates.
- Fiscal foreshadowing: a labor income tax taxes *the insurance technology of the wealth-poor*-.

Transition: From Hours to Participation

What Pijoan-Mas leaves open:

- His margin is *intensive*—continuous hours. But the labor markets facts say most cyclical movement in aggregate hours is *extensive*: bodies moving between employment and non-employment.
- An extensive margin means a **discrete choice**, and discrete choices aggregate very differently from smooth ones: the distribution's role becomes even starker.
- The next three parts follow the extensive margin from its 1980s complete-markets treatment (Hansen–Rogerson) through its incomplete-markets repair (time averaging) to its stochastic GE quantification (Chang–Kim).

The Tension the Extensive Margin Must Resolve

- Micro evidence: intensive-margin elasticities for prime-age workers are small—roughly 0 to 0.5.
- Macro practice: business-cycle models need aggregate elasticities of 1 to 4 to match employment volatility.
- An order-of-magnitude gap, and slide 10's units point (Frisch vs. Marshallian) closes only part of it.
- The rest of the resolution runs through **indivisibility**: if individuals choose employment, not hours, then the aggregate elasticity is a property of *who is near the participation margin*—a distributional object with no micro counterpart.

Indivisible Labor as a Fact

- Most workers face discrete choices: a job is 40 hours or it is not held; part-time work is a different job, not a point on a continuum.
- In the data: the variance of annual hours across workers is dominated by weeks worked, not hours per week; cyclical movements in aggregate hours are two-thirds-plus bodies.
- So model it directly: $h \in \{0, \bar{h}\}$, with a fixed utility cost $\bar{B}$ of working.
- The question becomes: who works, and how does that mass respond to prices?

The Discrete Choice

Each period the household compares two continuation problems:

$$V(a, e) = \max\left\{V^E(a, e),\ V^N(a, e)\right\},$$

where V^E includes earnings $we\bar{h}$ and the utility cost $\bar{B}$, V^N includes neither, and both contain the saving choice.

- The employment decision is an **option on one's own productivity**, exercised subject to wealth.
- The optimal policy, we now show, is a cutoff rule.

Why a Cutoff, and Why It Rises with Wealth

Let $\Delta(a, e) \equiv V^E(a, e) - V^N(a, e)$, the gain from working.

- Δ is **increasing in** e: earnings scale with productivity while the utility cost does not $\Rightarrow$ a reservation rule, work iff $e \geq e^*(a)$.
- Δ is **decreasing in** a: wealth raises consumption and lowers marginal utility, shrinking the value of the same earnings against the same cost $\Rightarrow$ e^* is increasing in a.
- In words: *the wealthy can afford to be choosy about when they work*.

The Threshold, Exactly (Static Toy)

Log utility, no saving: indifference means the *proportional* consumption gain compensates the fixed cost,

$$\ln\big((1+r)a + we\bar{h}\big) - \bar{B} = \ln\big((1+r)a\big) \quad \Longrightarrow \quad e^*(a) = \frac{(1+r)\,a\,(e^{\bar{B}}-1)}{w\bar{h}}.$$

- At $\bar{B} = 0.3$, $\bar{h} = 1/3$, $w = 1$, $r = 0.04$: $e^*(a) \approx 1.09\,a$.
- Same earnings, different proportional gain: large to the poor, small to the rich.

Hansen–Rogerson: the Lottery Solution

The 1980s faced a technical problem: indivisibility makes the choice set nonconvex, and a representative agent cannot "work a bit less" in a recession.

- Hansen (1985) and Rogerson (1988) restored convexity with **employment lotteries**: households choose a *probability* n of working $\bar{h}$, and complete markets insure the draw—consumption is C whether you work or not.
- Elegant, tractable, and—as the next slide shows—it manufactures an extreme answer to the elasticity question.

The Lottery Derivation, in Two Lines

With the draw fully insured, expected utility is

$$\ln C \;-\; n\bar{B} \qquad \text{— } \textbf{linear} \text{ in } n.$$

- Whatever curvature individuals had, the stand-in household's disutility of aggregate employment is linear: its Frisch elasticity is **infinite**, and the hours condition degenerates to $w\bar{h} = \bar{B}\,C$.
- This is how early RBC reconciled small micro elasticities with large fluctuations: the elasticity was manufactured by the lottery market, *not by preferences*.
- The mechanism is real (extensive margin); the insurance is fictional (nobody trades employment lotteries). The question is what survives when the insurance is honest.

Time Averaging

Ljungqvist and Sargent's fix: drop the lotteries, keep indivisible labor, put the household in a Bewley economy.

- Without insurance markets, a household smooths the employment event *across its own work history*: work some periods, live off the buffer stock in others—**time averaging**.
- The buffer stock is imperfect insurance, so the stand-in linearity breaks: the infinite elasticity is tamed to a *large, finite* one.
- And crucially, how large depends on who sits near the thresholds—the elasticity becomes **distribution-dependent**.

Summary Table

	smoothing device	**agg. elasticity**	**who bears risk**
Hansen–Rogerson	lotteries (complete mkts)	infinite	nobody
time averaging	own buffer stock	large, finite	the unlucky
Chang–Kim	buffer stock, stoch. GE	≈ 1	the wealth-poor

- Common mechanism: extensive margin + *some* smoothing. What differs is *which* smoothing—and that matters mostly for distribution and welfare, which is why we care about here.

Chang–Kim (2007): the Question

"Heterogeneity and Aggregation: Implications for Labor-Market Fluctuations" (*AER*).

- When employment decisions come from a cross-sectional distribution, what does aggregate labor supply look like over the business cycle? Two sub-questions:
- **(1)** What aggregate labor supply elasticity *emerges* from self-insured indivisible labor—given that nothing forces it to equal anything?
- **(2)** If an econometrician fits a representative-agent model to this economy's aggregates, what are the findings?
- It is the stochastic GE resolution of the micro–macro dispute.

Environment

A Krusell–Smith economy with the Part 3 labor margin:

- Households: one riskless asset, borrowing limit, idiosyncratic productivity from a persistent AR(1) estimated on PSID wages, indivisible labor $h \in \{0, \bar{h}\}$—no lotteries; self-insurance only.
- Firms: aggregate production in capital and effective labor, hit by aggregate TFP shocks.
- Aggregate state: TFP and the wealth–productivity distribution λ; **approximate aggregation** (a small set of moments forecasts prices) makes it computable —and holds here, as in Krusell–Smith.
- The reservation rule now shifts with the cycle: a boom (higher w) lowers e^* at every wealth level, pulling marginal households into work.

Aggregation, in One Line

Aggregate employment is the mass above the thresholds, $N = \int \mathbf{1}\{e \geq e^*(a)\}\, d\lambda$, so

$$\frac{\partial N}{\partial \ln w} = -\int \varphi\big(e^*(a) \,\big|\, a\big)\, \frac{\partial e^*(a)}{\partial \ln w}\, d\lambda_a \;>\; 0.$$

- An integral of **densities of households at their margins**, weighted by threshold slopes.
- No preference parameter appears except through the slope of e^*: the aggregate elasticity is an *equilibrium, distributional* object.

Calibration Discipline

- The idiosyncratic process comes from panel wage data , *not* from matching aggregate moments—so the distribution of who-is-near-the-margin is disciplined independently of the answer.
- The *individual* (intensive-margin) elasticity is set small, in the micro range: **the paper does not assume its conclusion**.
- Wealth distribution and average employment matched; aggregate TFP standard.
- Whatever aggregate elasticity comes out is then a genuine prediction of the mechanism.

Result 1: the Elasticity that Emerges

- Simulated aggregate hours and wages imply an aggregate labor supply elasticity of **around one**: several times the assumed individual elasticity, far below the lotteries' infinity—the middle row of our summary table, quantified.
- The amplification is manufactured by *threshold crossings*, not preference curvature: marginal households flowing across e^* as prices move.
- Verdict on the micro–macro fight: both sides measured real objects; they measured **different** objects. Micro elasticities describe workers far from thresholds; the aggregate is dominated by those near them.

Result 2: Who Carries the Fluctuations

- The households who switch employment status over the cycle are those near their thresholds: predominantly **low-wealth households with middling productivity**.
- The wealthy barely respond (thresholds far above their draws); the very poor work almost always (thresholds near zero).
- So employment volatility is carried by an identifiable, *changing* subpopulation—which is why the aggregate elasticity moves when the wealth distribution moves, and why no fixed parameter can summarize it.
- A testable cross-sectional prediction: employment cyclicality concentrates where the model says the margin lives.

Result 3: the RA Simulation

The paper's most-quoted experiment: fit a representative-agent model to this economy's *simulated aggregates*.

- The stand-in household's intratemporal condition (MRS = wage) fails period by period in the aggregated data: reconciling hours, consumption, and wages requires residual **"preference shocks."**
- But nothing shocked preferences. The residuals are **aggregation error**: the moving distribution masquerading as taste shifts.

Why the Elasticity Is Not a Primitive

- The integral a couple of slides ago depends on the joint wealth–productivity distribution: **anything that reshapes λ reshapes the elasticity**.
- Fiscal experiments reshape λ by design—taxes, transfers, debt . Evaluating them with a fixed "macro elasticity" plugs a moving target in as a constant.
- Same lesson as Pijoan-Mas at the other margin: there, *hours* respond to wealth; here, *participation* does. Both say the labor-supply block of a macro model is an equilibrium outcome, not a parameter to look up.

Transition: From Quantities to Prices

What the story so far holds fixed:

- Parts 1–4 studied the *supply* of labor—hours and bodies—against a technology where capital and labor are generic aggregates and one wage prices efficiency units.
- But capital interacts with labor on the **technology side** too, and not symmetrically across workers: some labor is complemented by machines, some is substituted.
- That asymmetry moves *relative wages*—the skill premium—and hence the earnings distribution.
- The definitive quantitative treatment: **KORV (2000)**.

KORV: a Different Kind of Capital–Labor Question

Krusell, Ohanian, Ríos-Rull and Violante (2000), *Econometrica*.

- So far capital entered as a savings vehicle and a factor paying one wage per efficiency unit. KORV ask about capital's *composition effect on the wage structure*: which workers do machines complement?
- Their answer organizes the biggest wage fact of the postwar era—and hands this course a technology-side theory of the skill premium that statistical processes only describe.

The Fact to Explain

- The **skill premium** (college/high-school wage ratio) fell into the 1970s, then rose sharply from about 1980—*while the relative supply of skilled workers grew throughout*.
- With a standard two-factor CES, rising relative supply should depress the premium. Something shifted relative demand for skill, persistently and strongly.
- The 1990s label was "skill-biased technical change"— correct but, in the Katz–Murphy tradition, a *residual*: an unmeasured trend in the production function's taste for skill, growing at whatever rate fits.

The Idea: Measure the Bias, Do Not Assume It

- KORV replace the residual with a **measured** force: the postwar collapse in the quality-adjusted price of *equipment*, and the capital deepening it induced.
- One assumption converts equipment growth into demand for skill: **capital–skill complementarity**—equipment substitutes for unskilled labor but complements skilled labor (Griliches's hypothesis, made quantitative).
- Then "skill-biased technical change" is investment-specific technical change, priced in observable markets, working through the production function's substitution structure.

The Production Function

Structures k_s separable; a nested CES over equipment k_e, skilled labor s, unskilled labor u:

$$y = k_s^{\alpha}\Big[\mu\, u^{\sigma} + (1-\mu)\big(\lambda\, k_e^{\rho} + (1-\lambda)\, s^{\rho}\big)^{\sigma/\rho}\Big]^{(1-\alpha)/\sigma}.$$

- Equipment sits *inside a nest with skilled labor*; unskilled labor sits outside.
- Elasticities of substitution: $1/(1-\sigma)$ between the nest and unskilled labor; $1/(1-\rho)$ within the nest.
- **Capital–skill complementarity** $\equiv \sigma > \rho$. Whether it holds is what the data decide.

Deriving the Premium (I): Marginal Products

Write $G \equiv \lambda k_e^{\rho} + (1-\lambda)s^{\rho}$. Differentiating:

$$w_u = y^{1-\sigma} A^{\sigma} \mu \, u^{\sigma-1}, \qquad w_s = y^{1-\sigma} A^{\sigma} (1-\mu) \, G^{\sigma/\rho-1} (1-\lambda) \, s^{\rho-1},$$

(common factors collected in A).

- Take the ratio: the common factor cancels—the premium depends only on the *relative* structure of the nest.

Deriving the Premium (II): the Equation

Factor $G = s^{\rho}\left[\lambda(k_e/s)^{\rho} + 1 - \lambda\right]$ and collect the $\ln s$ terms:

$$\ln \frac{w_s}{w_u} = \text{const} + (1-\sigma)\ln\frac{u}{s} + \frac{\sigma-\rho}{\rho}\ln\left[\lambda\left(k_e/s\right)^{\rho} + 1 - \lambda\right].$$

- Two observable forces: relative supply u/s and equipment per skilled worker k_e/s.

The Decomposition in Changes

Differentiating:

$$d \ln \frac{w_s}{w_u} = -(1-\sigma)\, d \ln \frac{s}{u} + (\sigma - \rho)\, \omega\, d \ln \frac{k_e}{s}, \qquad \omega = \frac{\lambda (k_e/s)^{\rho}}{\lambda (k_e/s)^{\rho} + 1 - \lambda}.$$

- A **relative-supply effect** (depresses the premium, strength $1-\sigma$) against a **capital-deepening effect** (raises it, strength $(\sigma-\rho)\,\omega$, with ω the equipment share within the nest).
- $\sigma = \rho$ collapses the nest: no complementarity, no deepening effect, and the premium can only fall as supply grows. The whole account lives in $\sigma - \rho > 0$.

The Crucial Input: Equipment Prices

- The deepening series k_e/s is built from investment deflated by the **quality-adjusted relative price of equipment** (Gordon's series and successors): a computer in 1990 is many 1970 computers.
- That price fell dramatically and persistently, so measured equipment stocks grew far faster than nominal investment suggests.
- This is the paper's discipline: the force driving the premium is *priced in a market*, not inferred from the premium itself. Labor quantities and wage bills come from the CPS; series construction is half the paper's work.

Estimation and Identification

- $(\sigma, \rho, \lambda, \mu, \dots)$ estimated by on annual U.S. aggregates; measurement equations for the **labor share** and the **skill premium**; latent capital utilization absorbs cyclical mismeasurement.
- Identification, read off the decomposition: the premium–supply comovement reveals $1 - \sigma$; the premium–deepening comovement reveals $(\sigma - \rho)\,\omega$.
- Nothing about the premium's *trend* is imposed; fitting it is the test.

Estimates and the Fitted Account

- $\sigma \approx 0.40$, $\rho \approx -0.50$ (exact values and standard errors: their tables): substitution elasticities ≈ 1.67 equipment–unskilled, ≈ 0.67 equipment–skilled. **Complementarity confirmed.**
- The fitted series then tracks the premium's U-shape with no residual: college supply wins through the 1970s (premium falls); equipment deepening wins from $\approx$1980 (premium rises).

Critiques and Open Edges

- **Functional form carries weight**: the two-elasticity nest is the identifying restriction; aggregate time series cannot separate much richer substitution patterns.
- **Skill = college is coarse**: the 2000s polarization facts (growth at both tails) motivated task-based models where machines compete with *middle* tasks (Acemoglu–Autor).
- **Sample dependence**: extending through the 2000s–2020s stresses the fit as the premium plateaus while IT equipment keeps deepening.

Transition: Why KORV Belongs Here

- KORV is a complete-markets, aggregate-data paper—so why here? Because it closes a loop the household side needs: **the wage structure our households take as risk is itself produced by capital**.
- Descendants make the connection literal: Moll–Rachel–Restrepo (2022) put exactly this force inside a heterogeneous-agent economy, where automation moves *wealth* as well as wage inequality.

Two Literatures, One Missing Corner

- **DMP** : equilibrium unemployment from matching frictions—with *risk-neutral* workers, so unemployment risk carries no insurance problem.
- **Aiyagari / Krusell–Smith** (8002 + this topic): risk-averse households self-insuring—with a *frictionless* labor market, so the risk process is exogenous.
- Each side assumes away the other's central object. KMŞ build the missing corner: **precautionary savers whose income risk is generated by a matching market**, with aggregate shocks.
- The capstone by construction: every ingredient is something this course has built.

Why Force the Marriage?

Two questions, each only half-answered by one parent literature:

- **How much unemployment insurance?** In Bewley–Huggett–Aiyagari (BHA) the hazards are exogenous: UI is pure insurance, and full insurance is optimal. In risk-neutral DMP it is pure distortion: under Hosios, output is highest with no UI. Which force wins?
- The trade-off is **insurance vs. job creation**, not insurance vs. search effort: there is no moral hazard in the model.
- **How does productivity move unemployment?** Do the Shimer puzzle and the Hagedorn–Manovskii resolution survive risk aversion, precautionary saving, and capital?

What Earlier Attempts Gave Up

- **Merz (1995), Andolfatto (1996)**: DMP with risk aversion and capital—but idiosyncratic risk is fully insured. No precautionary motive, no distribution.
- **Imperfect-insurance versions** (Costain–Reiter, Bils–Chang–Kim, Nakajima, Shao–Silos, Rudanko, Jung–Kuester): each simplifies a hard piece—risk-neutral firm owners, union-style bargaining that removes wealth from the wage, or workers without assets.
- The hard pieces: (i) **wage determination** when the outside option depends on wealth; (ii) **firm valuation** with missing markets—whose discount factor prices a vacancy?
- KMŞ confront both and *nest both parents*: risk aversion $\to 0$ gives DMP; matching efficiency $\to \infty$ gives BHA.

Environment: Technology and Matching

- Discrete time, one period = six weeks. Measure one of consumers, each employed or unemployed.
- Each matched worker produces $zF(k)$ with rented capital k, which depreciates at rate δ.
- Matching function $M(u, v)$, tightness $\theta = v/u$, hazards $\lambda_f = M/v$ and $\lambda_w = \theta\lambda_f$, exogenous separation σ:

$$u' = \big(1 - \lambda_w(\theta)\big)\, u + \sigma(1 - u).$$

- **No labor-supply margin**: leisure is not valued and every offer is accepted. The margin of Parts 1–4 is switched off; employment moves only through job creation.
- Notation: the paper writes h for our b and ξ for our κ.

Asset Structure

- No insurance markets for employment risk. Two assets: **capital** k, rented to firms at r, and **equity** x, a claim on the *aggregate* profits of all jobs (price p, dividend d).
- Equity is what is new relative to Aiyagari: matched firms earn rents, and someone must own them. Claims on individual jobs are ruled out—a worker would short her own job.
- Without aggregate risk both assets are riskless, so no arbitrage requires

$$p = \frac{d + p}{1 + r - \delta}, \qquad q \equiv \frac{1}{1 + r - \delta}.$$

- Portfolio composition is then irrelevant; one state variable suffices: $a \equiv (1 + r - \delta)\, k + (p + d)\, x$.

The Household Side, Written Out

Steady state, borrowing limit $a' \geq -\underline{a}$:

$$\tilde{W}(w, a) = \max_{a'} \; u(c) + \beta\big[\sigma\, U(a') + (1-\sigma)\, W(a')\big], \qquad c + q\,a' = a + w,$$

$$U(a) = \max_{a'} \; u(c) + \beta\big[(1-\lambda_w)\, U(a') + \lambda_w\, W(a')\big], \qquad c + q\,a' = a + b,$$

with $W(a) \equiv \tilde{W}\big(\omega(a), a\big)$.

- $\tilde{W}$ is the value at an *arbitrary* wage w—the object bargaining needs. W imposes the equilibrium wage function $\omega(a)$, renegotiated every period. Policies: $\psi_e(a)$ and $\psi_u(a)$.
- The transition probabilities are **equilibrium DMP hazards**, not an exogenous chain: household risk is an endogenous object.

Firms: Vacancies and Filled Jobs

$$V = -\kappa + q\Big[(1-\lambda_f)\,V + \lambda_f \int J\big(\psi_u(a)\big)\,\frac{f_u(a)}{u}\,da\Big],$$

$$\tilde{J}(w,a) = \max_k \; zF(k) - rk - w + q\Big[\sigma V + (1-\sigma)\,J\big(\tilde{\psi}_e(w,a)\big)\Big], \qquad J(a) \equiv \tilde{J}\big(\omega(a),a\big).$$

- Capital choice is static: $r = zF'(k)$, the same k at every job, so $\tilde{k} = \bar{k}/(1-u)$.
- **Free entry**: vacancies are posted until $V = 0$.
- Firms discount at q: the market rate, and the marginal rate of substitution of every unconstrained shareholder—so owners agree on what the firm should do.

What Is New in the Firm's Problem

- J **depends on the worker's wealth.** The wage $\omega(a)$ does, and wealth drifts along ψ_e while the match lasts: the firm must forecast its worker's saving.
- **The distribution enters free entry.** A vacancy meets a random draw from the unemployed, density $f_u(a)/u$. A poorer pool of searchers is a more profitable pool.
- Dividends aggregate flow profits net of recruiting:

$$d = \int \pi(a) f_e(a)\, da - \kappa v, \qquad \pi(a) = zF(\tilde{k}) - r\tilde{k} - \omega(a).$$

- And $p + d = \int J(a) f_e(a)\, da$: **the stock market is the sum of match values**. Hold that thought.

Where Integration Bites: the Wage Bargain

- Under Nash bargaining, the wage splits surplus relative to *outside options*—and a worker's outside option now depends on her **wealth**: a poor worker cannot afford unemployment, so her threat is weak.
- In principle, then: wages are wealth-specific, $\omega(a)$; firms care *whom* they match with; the value of a vacancy depends on the distribution of searchers' wealth; the distribution enters the bargaining problem itself, not just market clearing.
- This is the conceptual price of the marriage—and handling it carefully, rather than assuming it away, is a large part of the paper's contribution.

Nash Bargaining, Written Out

$$\omega(a) = \arg\max_{w} \left[\tilde{W}(w,a) - U(a)\right]^{\gamma} \left[\tilde{J}(w,a) - V\right]^{1-\gamma},$$

$$\text{FOC:} \quad \gamma \tilde{W}_w \left[\tilde{J}(w,a) - V\right] = (1-\gamma)\left[-\tilde{J}_w\right]\left[\tilde{W}(w,a) - U(a)\right].$$

- Risk neutrality: $\tilde{W}_w = -\tilde{J}_w = 1$. This is the constant surplus split seen before, and the wage does not depend on a.
- Here $\tilde{W}_w = u'(c)$ by the envelope theorem: the worker's surplus is in *utility* units, and both $u'(c)$ and $\tilde{W} - U$ vary with wealth.
- No commitment: wages are reset every period, so today's wage moves the worker's saving and tomorrow's bargain ($-\tilde{J}_w \neq 1$).

Why $\omega(a)$ Is Increasing and Concave

- **Increasing**: a low-wealth worker has a thin buffer against unemployment, so her outside option is worse and she settles for less.
- **Bounded above**: as $a \to \infty$ the worker is perfectly self-insured and bargains like a risk-neutral agent, so $\omega(a)$ approaches the DMP wage.
- Increasing and bounded: the schedule must flatten. The open issue is quantitative—*where* it flattens, and how many households sit on the steep part.
- If most sit on the flat part, the wage block is "DMP plus a correction near the constraint."
- In the data: wealth raises reservation wages (Alexopoulos–Gladden, SIPP).

Stationary Equilibrium

Values (W, U, J, V), policies (ψ_e, ψ_u), a wage function $\omega(a)$, prices (r, p, d), tightness θ, and distributions (f_e, f_u) such that:

1. Households optimize given $\omega(\cdot)$, r, and $\lambda_w(\theta)$.
2. Firms: $r = zF'(\tilde{k})$; J solves its Bellman equation; free entry, $V = 0$.
3. $\omega(a)$ solves the Nash problem at every a.
4. Asset market: $\int \psi_e(a) f_e(a)\, da + \int \psi_u(a) f_u(a)\, da = (1 + r - \delta)\,\bar{k} + p + d$.
5. Stationarity: $u = \sigma/(\sigma + \lambda_w)$, and (f_e, f_u) reproduce themselves.

Savings supply the capital that matched firms rent, so r and θ are determined *jointly*.

Calibration: Common Parameters

- Technology: $zF(k) = zk^{\alpha}$ with $\alpha = 0.3$, $\delta = 0.01$, $\beta = 0.995$ per six-week period. Targets: capital share 0.3, investment–output ratio 0.2, annual return on capital 4%.
- Preferences: $u(c) = \log c$ (CRRA with coefficient 5 as robustness). Borrowing limit $\underline{a} = 0$.
- Flows, from Shimer's monthly numbers: separation $\sigma = 0.05$ (monthly 0.034); job finding $1 - (1 - 0.45)^{1.5} \approx 0.59$ per period. Steady-state unemployment is then close to 7.7%.
- The contested parameter is b. It is hard to measure, the literature offers two strategies, and the paper stays agnostic: **every experiment is run under both**.

Two Labor-Market Calibrations

	Shimer (benchmark)	**Hagedorn–Manovskii**
unemployed income b	0.99 ($\approx$ 40% of mean wage)	2.29 ($\approx$ 96% of mean wage)
worker share γ	0.72 ($= \eta$, Hosios)	0.052
matching function	$\chi u^{\eta} v^{1-\eta}$, $\chi = 0.6$, $\eta = 0.72$	$uv/(u^{l} + v^{l})^{1/l}$, $l = 2.2$
tightness θ	1	0.7
vacancy cost κ	0.532	2.165
unemployment; mean wage	7.69%; 2.48	7.81%; 2.38

- Same aggregates, very different surplus: under HM the worker is nearly indifferent between working and not.

Result 1: Wealth Barely Moves the Wage

- Quantitatively, the bargained wage schedule is **nearly flat in wealth**: $\omega(a)$ rises noticeably only for assets very close to the borrowing constraint.
- Size: the mean-to-minimum wage ratio is 1.02 under the Shimer calibration and 1.0002 under HM. The feared $\omega(a)$ complication is real in principle, modest in practice.
- Why: self-insurance works well enough that, away from the very bottom, workers' effective exposure to the employment event is limited—surpluses and threat points move little with a for most of the distribution.
- Practical consequence: the wage block stays close to its risk-neutral counterpart—first hint of the aggregation result.

A By-Product: Very Little Dispersion

- Households differ only in employment histories. Spells are short (job finding near 0.6 per six weeks), so buffers are rarely run down and almost nobody sits at the constraint.
- Wages differ only because wealth does: a mean–min ratio of 1.02 is far below measured frictional wage dispersion (the Hornstein–Krusell–Violante yardstick).
- No ability, match-quality, or preference heterogeneity: the authors flag wealth and wage dispersion as the benchmark's empirical weak spot.
- Does the flat-wage result depend on it? They add discount-factor heterogeneity to spread wealth out and put far more mass on the steep part of ω. The aggregate responses are unchanged.

The UI Experiment: Setup

- Reinterpret b: no home production; all income of the unemployed is a government benefit, financed by a proportional tax τ with period-by-period budget balance:

$$c + qa' = a + (1-\tau)\, w, \qquad c + qa' = a + (1-\tau)\, b, \qquad \tau \int \omega(a) f_e(a)\, da = (1-\tau)\, u\, b.$$

- Shimer calibration, benchmark $b = 0.99$. Vary b from 0 to 1.5 and compare steady states.
- Raising b works through three channels: (1) **insurance**—consumption falls less at job loss; (2) **job creation**—a better outside option raises wages, lowers profits and vacancies, raises unemployment; (3) **capital**—less precautionary saving.
- Aiyagari has (1) and (3). Channel (2) is what DMP adds.

UI: Effects on Quantities

- The bargained wage rises with b, and so does the unemployment rate: fewer vacancies per searcher.
- Capital per worker falls, but only slightly: from 72.17 at $b = 0$ to 72.07 at $b = 1.5$. The precautionary motive is weak to begin with.
- **Steady-state output is highest with zero (or even negative) UI**—the risk-neutral Hosios logic carries over.
- But output is not welfare. Risk-averse households value the insurance, and higher-b steady states have less capital partly because households *need* less of it. We need a welfare measure that does not penalize UI for that.

The Welfare Measure

- Many "countries," identical except for b, each in its steady state. Move every consumer of the benchmark country ($b = 0.99$) to another one, *with her employment status and assets*.
- Her consumption-equivalent gain λ solves

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log\big((1+\lambda)\, c_t\big) = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log \tilde{c}_t \quad \Longrightarrow \quad \lambda = \exp\big((\tilde{V} - V)(1-\beta)\big) - 1.$$

- Aggregate with the benchmark's invariant distribution; report λ by group and the share with $\lambda > 0$.
- A mover keeps her wealth, so high-UI countries are not charged for their lower capital. It is *not* a transition path.

Welfare Effects of Changing UI

new b	all	unemployed	employed	share gaining	poorest unemp.	poorest emp.
0.01	0.11%	−0.09%	0.13%	92.10%	−4.0%	−1.0%
0.25	0.12%	−0.02%	0.14%	92.29%	−1.2%	−0.3%
0.50	0.11%	0.01%	0.12%	99.22%	−0.54%	−0.09%
0.75	0.07%	0.02%	0.07%	99.96%	−0.2%	0.00%
1.25	−0.14%	−0.09%	−0.15%	0.00%	0.06%	−0.1%
1.50	−0.38%	−0.27%	−0.39%	0.00%	−0.04%	−0.32%

Average λ from moving a benchmark ($b = 0.99$) consumer to the steady state with the new b. Source: KMŞ (2010), Table 2, Shimer calibration.

Reading the Table

- **Less UI is preferred by almost everyone.** Cutting b from 0.99 to 0.75 benefits 99.96% of households; only the borrowing-constrained unemployed lose.
- The average gain peaks near $b \approx 0.30$—about 12% of the mean wage, against 40% in the benchmark. Below that, lost insurance and lower wages take over.
- **More UI is worse for essentially everyone**: at $b = 1.5$ the mean loss is 0.38%.
- Magnitudes are lopsided: mean effects near 0.1% of consumption, but -4% for the poorest unemployed when UI is removed. Job creation dominates on average; insurance matters for a group the average barely sees.

The Same Experiment in Aiyagari

- Freeze the hazards at their benchmark values (job finding 0.6, separation 0.05): unemployment is 7.69% whatever b is, and UI and its tax distort nothing.
- Then welfare of the unemployed is *monotonically increasing* in b, the employed are nearly indifferent, and full insurance ($b = w$) is Pareto efficient.
- **The policy prescription flips.** A planner using average λ wants much more UI in the Aiyagari economy and less in KMŞ—same households, same risk at the benchmark, same market incompleteness.
- The lesson for the fiscal topics ahead: an income process that is exogenous to policy is a Lucas-critique problem. If policy moves the risk, the welfare answer can change sign.